School of Computer Science
FACULTY OF ENGINEERING AND
PHYSICAL SCIENCES

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COMP5122M Data Science

Coursework 2

Global Climate Attitudes and Eco-Anxiety — Part II

Group Gloom

Yasemin Bal — sc22y2b
Evan Madurai — sc22em
George Hester — sc22gh
Andreas Bazigos — sc222ab

Predictive Modelling

The goal of this model is to predict whether a given demographic of individuals holding a certain set of attitudes are in support of strengthening climate commitments. In particular, the aim is to discern the strongest determinant of this sentiment between demographic features and attitudinal features.

Data Processing and Classification Methodology

Data processing for this project was completed in Python using `numpy` and `pandas` for the initial data manipulation to generate the training data, `sklearn` to produce a model using regularised logistic regression (LR) and a random forest (RF) classifier. Results were then visualised using `matplotlib` and `seaborn` to proceed with analysis.

Country-level breakdowns of the responses are available for 73 countries, with Kazakhstan, the Netherlands, Poland and Sweden included only in global estimates due to small country samples sizes.
Nine of the sampled countries were classified as Small Island Developing States (SIDS) and nineteen were classified as Least Developed Countries (LDCs). These categories provide insight into grouped demographical information, therefore, a more extensive country classification system was developed that groups Global, LDCs, SIDSs, both an LDC and SIDS and Other. These groupings provide another demographic feature for the model and enable regional analysis for later clustering.

Data Standardisation

Columns were renamed to a more human-readable format, with country names undergoing a standardisation process. For example, formerly "DR Congo" emerged as "Democratic Republic of the Congo", in an effort to forgo brevity and enhance both documentation and clarity.

Target Variable Isolation

The dataset was filtered to retain responses from the survey question: "Should your country strengthen or weaken its commitments to address climate change?" From these responses, the target variable: "strengthen" was identified as the key metric for sentiment evaluation. These targets directly address the core research objective and the response options create a clear binary classification problem post-processing.

Supervised Learning Preparation

The dataset contains aggregated percentages rather than individual responses. Therein, each row represents the percentage of a specific demographic group within a country who have selected a particular response option. These percentages represent attitudinal data points and were transformed into a model-digestible format. To do this, the response "Don't know" was removed, as it can be treated as a lack of opinion and subsequently excluded due to the attitude predictions explicitly demanding opinion. After this, each question and response pair was pivoted into an individual column, leaving a single row per demographic containing its attitudinal data.
From this, a binary target variable was constructed. Continuous percentages can be converted to a binary classification by enforcing a threshold to categorise responses. A temporary mean made from responses in strong favour or "Strengthen" subtracted by those opposed or "Weaken" was compared with the median to categorise responses into either binary category. This median net-support threshold was chosen as it is less influenced by outliers, unlike the mean. For example, a country with 80% Strengthen, 10% Weaken has stronger support than 50% Strengthen, 5% Weaken; this approach enables predictions above or below the median support level.

Model Pipeline Construction

The data was split into a standard 70-30% training-testing split, a standard practice for evaluating model generalisation as it provides data which can be used to evaluate model performance. Furthermore, a random pattern was applied during the split, mitigating any over-fitting the model may perform for specific demographics. This behaves as a held-out test set [3], avoiding bias during evaluation and ensuring the model performs accurately on unseen data, like in the testing set. Additionally, a random state was set to ensure reproducibility.

Modelling Methodology

To prepare the dataset for model training, `sklearn` required a particular setup. Numerical values needed little manipulation as they were accepted by the model, unchanged; undefined values were set to the mean of their surrounding data; and categorical values such as age or education, were reshaped through the use of one-hot encoding [6] where new binary columns were created for each possible value of the categorical data, ensuring the model accepts strictly numerical inputs across the entire dataset.

Hyper-parameter Optimisation

Before training began on the entire training set, hyper-parameter optimisation [2] was performed. This approach uses a cross-validated grid-search, with the evaluation method set to F1 accuracy for measuring model performance. The following parameters were then chosen for regularised logistic regression.

- **C** - The inverse of the regularisation strength, this value is used to prevent over-fitting where a lower value forces the model to not fit to the training data as closely.
- **Penalty** - Determines the type of regularisation to apply, `l1` is likely to eliminate less important features and `l2` would spread weightings across all features.
- **Solver** - Determines the algorithm to be used to optimise the objective function and approach the target value.
- **Maximum Iterations** - The number of iterations that the algorithm can run, lower values may cause the model to not converge.

This optimisation was also performed on random forest using the following parameters.

- **Number Of Estimators** - The number of decision trees, where each tree is trained on a subset of the overall training set. More trees generally leads to a more robust and stable model when introducing new data.
- **Maximum Depth** - The maximum depth each decision tree will reach.
- **Minimum Samples Split** - The number of data points a node must have before it can be split further.
- **Minimum Samples Leaf** - The number of data points each leaf node in the tree must have.
- **Maximum Features** - The maximum number of features the algorithm will consider when looking at the best split for a given node based on the overall number of features provided.

Logistic Regression		**Random Forest**	
Parameter	Best Value	Parameter	Best Value
C	0.01	Number Of Estimators	100
Maximum Iterations	5000	Maximum Depth	20
Penalty	`l2`	Minimum Samples Split	2
Solver	`saga`	Minimum Samples Leaf	1
		Maximum Features	`log2`
F1 Accuracy	0.7479	F1 Accuracy	0.8464

Figure 1: Results from hyper-parameter optimisation.

The F1 accuracy shown in Figure 1 is based on the subset of the training data provided to the model by the *grid-search*. Because of this, results will not match after training has been performed on the entire training set.

Model Training and Evaluation

To perform model training, new instances of the logistic regression model and random forest classifier were created with the optimised hyper-parameters. These models were then provided the complete training set. After fitting, the testing set was used to predict target values and compared with the actual training set's target values using `accuracy_score` and `f1_score` functions from the `sklearn.metrics` package.

Importances features were extracted from the logistic regression classifier and the feature coefficients from the random forest classifier, highlighting which features most influenced the results of the classifiers. The results are shown in Figure 2.

Model Interpretation

Feature importance was extracted from both the random forest and logistic regression models to discern the weighting of demographics versus attitudes. This comparison of coefficient magnitude and impurity reduction enables the querying of feature importance by specific type, providing vital insight towards the target question.

Demographic vs Attitudinal Analysis

Models were separated and feature importance by type was extracted. Both models found *attitudinal* features are of **greater importance** than demographic importance as shown in Figure 3.

Specific Feature Analysis

Model interpretability techniques revealed a consistent hierarchy of predictive features. The strongest signals came from specific attitudinal responses, while aggregated demographic variables showed negligible importance (`Country Type [LDC]` scoring a low 0.52%).

Clustering analysis

In order to better understand how geography impacts people's attitude to climate change, a clustering analysis was conducted. We aggregated responses to similar climate questions into categories, conducted an unsupervised machine learning method, and compared the results across global regions.

Logistic Regression		**Random Forest**	
F1 Accuracy	0.83	F1 Accuracy	0.85
`sklearn` Accuracy	0.83	`sklearn` Accuracy	0.85
Top 5 Feature Coefficient (Absolute)		**Top 5 Feature Importance**	
Question 12 [Response 2]	-0.1330	Question 12 [Response 2]	0.0537
Question 3 [Response 4]	-0.1011	Question 12 [Response 1]	0.0396
Question 7 [Response 1]	-0.0999	Question 10 [Response 4]	0.0380
Question 8 [Response 2]	-0.0984	Question 11 [Response 2]	0.0296
Question 7 [Response 4]	-0.0978	Question 11 [Response 4]	0.0294
Coefficient By Type (Absolute)		**Importance By Type**	
Demographic	0.1336	Demographic	0.0518
Attitudinal	0.8664	Attitudinal	0.9482
		Top 5 Demographic Feature Importance	
		Country Type [LDC]	0.0052
		Education [Under 12]	0.0032
		Age [60 plus]	0.0029
		Country Type [SIDS]	0.0027
		Country [Mozambique]	0.0022

Figure 2: Results model training.

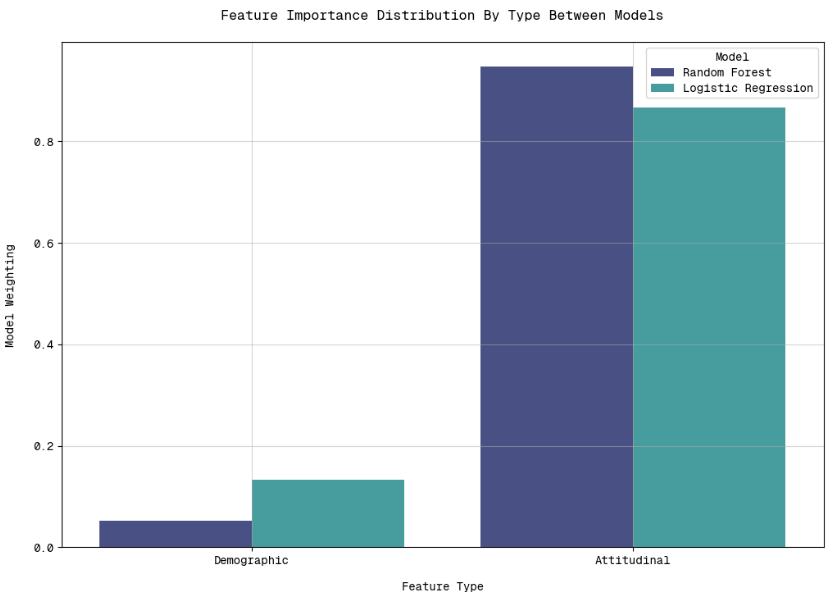

Figure 3: Results of the model weighting by feature type.

The most influential features in both models were responses to questions assessing institutional performance and climate risk perception. For instance, the response "About the same as now" to Question 12 (protecting people from extreme weather) was a top predictor, suggesting that *current levels of satisfaction or dissatisfaction with institutional protection* are a primary indicator of support for stronger climate commitments. Similarly, strong evaluations of business performance on climate (Question 7; responses "Very well" and "Very badly") and stances on nature protection (Question 11) were highly ranked. This pattern indicates that support is most strongly linked to an individual's *evaluation of climate risks and institutional responses* or broader *psycho-social attitudes* [1], rather than to demographic background.

Preprocessing

To bypass curse of dimensionality[1] not all 15 questions were included in the feature vector. Following a common rule of thumb in unsupervised learning that recommends approximately one feature per ten observations, 7 questions were chosen for 77 countries. Questions were chosen to represent the key dimensions of climate attitude, as seen in below.

Category	Question 1	Question 2
Eco-anxiety	Thinking about climate change	Worry for next generation
Policy preferences	Renewable energy transition	Protect and restore nature
Responsibility attribution	Country performance	Big businesses performance
Commitment support	Strengthening commitments	–

Pre-processing steps included removing aggregate "Global" entries to focus exclusively on country-level observations and filtering the data to include only aggregate demographic groups ("All education" and "All ages"). Responses to each question were rescaled to the range $[0, 1]$, where higher values indicate greater concern or support

[1]Phenomenon where as the number of features (dimensions) increases, data become increasingly sparse and distances between points lose meaning.

for climate action. “Don’t know” responses were excluded and the remaining response weights were renormalised to sum to one within each country–question pair. Finally, a composite score for each country and question was calculated by multiplying each response weight by its corresponding response score and summing across response categories.

K-means analysis

The unsupervised machine learning method chosen to analyse this data was K-means clustering. This was chosen due to it’s simplicity and scalability to large datasets. Additionally, it is also an easier method to interpret, since the resulting centroids represent typical cluster profiles.

We carried out an elbow test, which revealed a sharper decrease in significance at a cluster count of 5. Therefore it was decided to separate the data into 5 clusters representing 5 distinct categories of countries.

Cluster	Description	Examples
0	Countries with reasonably low responsibility attribution scores and high policy preferences scores.	Ethiopia, Pakistan, Bangladesh
1	Countries where opinions skew towards caring more about country performance than business performance; also among those with the highest eco-anxiety ratings.	Nigeria, Indonesia, India
2	Countries with quite low attitudes towards the responsibility attribution category.	Guatemala, Fiji, Algeria
3	Countries with the lowest attitudes towards the commitment support category.	USA, Canada, Australia
4	Countries with low eco-anxiety scores but similar policy preferences and responsibility attribution scores to other clusters.	Greece, Italy, Spain

Table 1: Cluster descriptions.

A Principal component analysis (PCA) was also conducted to attempt to visualise our results. We reduced our dimensionality to 2 principle components so it could be represented on an X and Y axis. This retained 55% of our variance, which is low but not surprising for human opinion survey data.

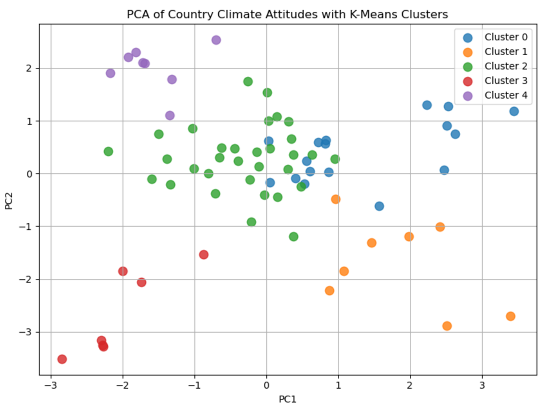

Figure 4: PCA projection onto two principal components

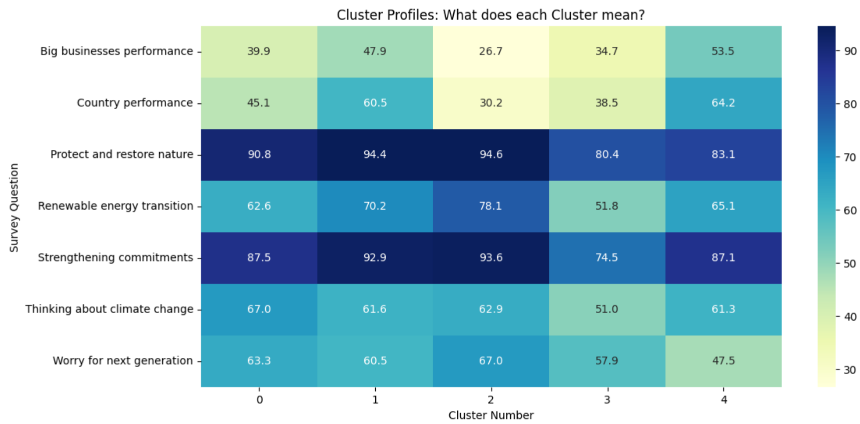

Figure 5: Heatmap representing the composition of the created clusters

Conclusions

From our analysis it was found that, the difference in age was not found to be a strong indicator of a prevalent, consistent age gap across countries in support for the renewable energy transition, instead, attitudinal factors overwhelmingly dominate demographic characteristics in determining this support. Various reasons could suggest this lack of prevalence like influence of age, which is often channeled through other variables. For example, younger people may support renewables more strongly partly because they are more educated on climate science and digitally literate [4], or more exposed to climate content on social media [5]. Crucially, demographic features like `Age Id` may not accurately capture these metrics and are better expressed through questions regarding attitude as age is not a monolithic predictor and therefore searching for a linear “age gap’ may be misplaced. When considering geographical regions of a country, we found they had effectively no correlation with the climate-change attitude category of that country. For example, within Asia countries were spread across all five categories with no clear majority. This lack of regional clustering suggests that geographical proximity does not strongly influence public attitudes toward climate change. Instead, national level factors such as political context, economic development, educational quality, and cultural values are likely to play a more significant role in shaping these attitudes. For example, Australia placed much closer in category to the USA and Canada than Fiji or Samoa.

Appendix

[1] Francesca Brambati, Daniele Ruscio, Flora Biassoni, Rian Hueting, and Anna Tedeschi. Predicting acceptance and adoption of renewable energy community solutions: the prosumer psychology. *Open Research Europe*, 2:115, 2022. Accessed: 2025-03-15.

[2] D2L. *What Is Hyperparameter Optimization?* 2023. Accessed: 2025-12-12.

[3] Kailaash Devi. Understanding hold-out methods for training machine learning models. 2023. Accessed: 2025-12-12.

[4] Energy Consumers Australia. Age is just a number ... until you bring up the energy transition, July 2023. Accessed: 2025-03-15.

[5] Cary Funk. Key findings: How americans' attitudes about climate change differ by generation, party and other factors, May 2021. Accessed: 2025-03-15.

[6] Wikipedia. One hot. 2025. Accessed: 2025-12-12.